

SRT210

DNS Servers
Secondary servers

Outline

- Configuring Secondary DNS with BIND
- Options for communication between primary and secondary servers.

Introduction

- Having only one DNS server introduces a single point of failure into your network.
- If this service fails, many other services no longer work.
 - Anything that depends on hostnames (e.g. your email server, AD, etc.).
 - Remember, most users will have trouble remembering (or just don't know) the addresses of machines they use.

Secondary Servers

- A second (or more) DNS server can be added to hold a duplicate copy of the zone data.
- Note that the communication between the primary and secondary will take place using TCP.

Terminology

- Primary and Secondary
 - AKA Master and Slave
 - A primary server holds a copy of the answers it can provide on the local disk.
 - An admin can manually change these.
 - Secondary servers get copies of the data from the primary.
 - And must not be manually changed.
 - This spreads the load across multiple machines and protects you in case one machine fails.
 - Any zone must have at least one primary, and should have more than one secondary.
 - One of which should be off-site.

Configuring DNS - Zones

- Each zone entry defines one zone that this server will be a name server for.
- At minimum:

```
zone "domain name" {  
    type secondary;  
    file <path to zone file>;  
    primaries {address list};  
};
```

Configuring Slave DNS - Zones

- Named has a directory of specifically for zone files.
 - `/var/cache/bind`
 - Note this may differ on other variants of Linux
- If you try to put a secondary file anywhere else, AppArmor may not let you.
 - Leave AppArmor on, and just put the secondary zones in the correct directory.
 - We'll also use it for dynamic DNS, so just get used to putting your zone files there.

Configuring Secondary DNS

- You can still use the parameters you have already learned.
- You can make those parameters different on the secondary than on the primary.
 - e.g. Maybe one is recursive, the other isn't.
 - e.g. Maybe one will answer queries from anyone, but the other will only answer to the local network.

Configuring DNS - notify

- notify “yes | explicit | no” - Whether or not to automatically notify other name servers in this zone when the data has changed.
 - yes will notify all other name servers in this zone.
 - Other servers to notify determined by NS records.
 - Note that a server will not notify itself, or the primary listed in the SOA record.
 - no will not notify anyone.
 - Useful on secondary servers.
 - explicit – Will only notify servers listed in the also-notify statement.

Configuring DNS - also-notify

- also-notify {address list;}; – A list of other servers to be notified when this server's data changes.
 - When notify is set to explicit, these are the only servers that will be sent notifications.
 - When notify is set to yes, these are in addition to the ones in the NS records,

Configuring DNS - allow-notify

- allow-notify {address list;}; - A list of servers that are allowed tell this machine there are updates. The primary is always allowed to, this allows you to specify others.
 - e.g. to create an hierarchy of multiple levels of secondary servers. The primary notifies a few secondaries (first-tier), they notify more secondaries (second-tier) and so on.

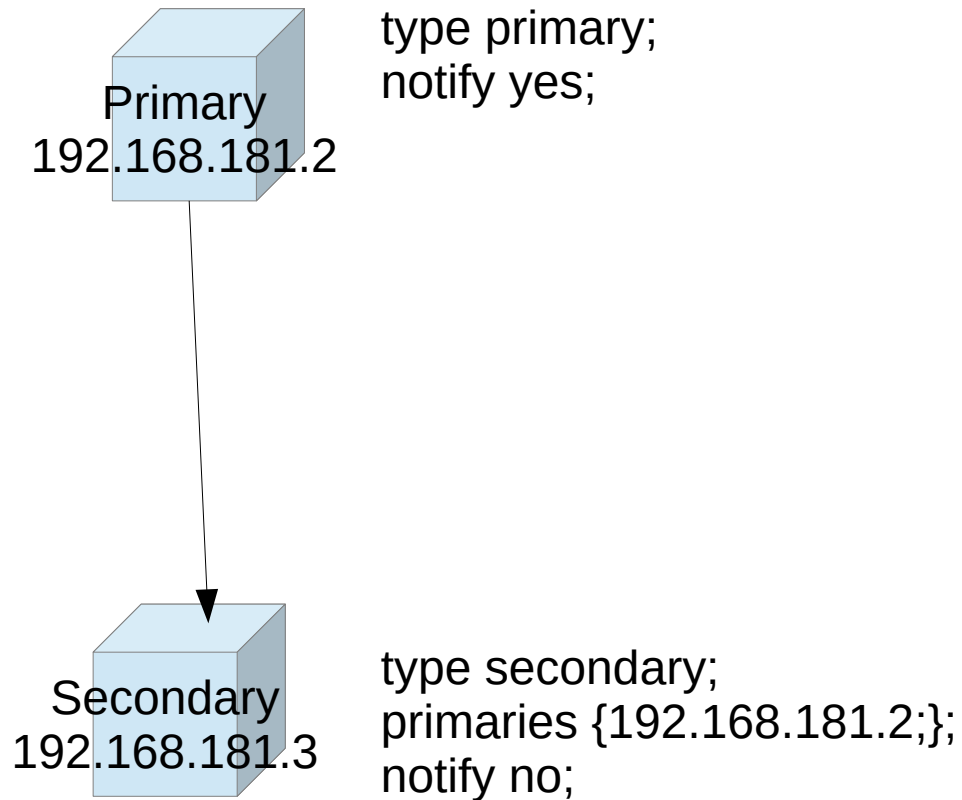
Configuring DNS - allow-transfer

- `allow-transfer {address list;};` - The list of servers that are allowed to request zone transfers.
 - By default, anyone can, but in a more secure environment, you might limit this by IP address, or by TSIG key.

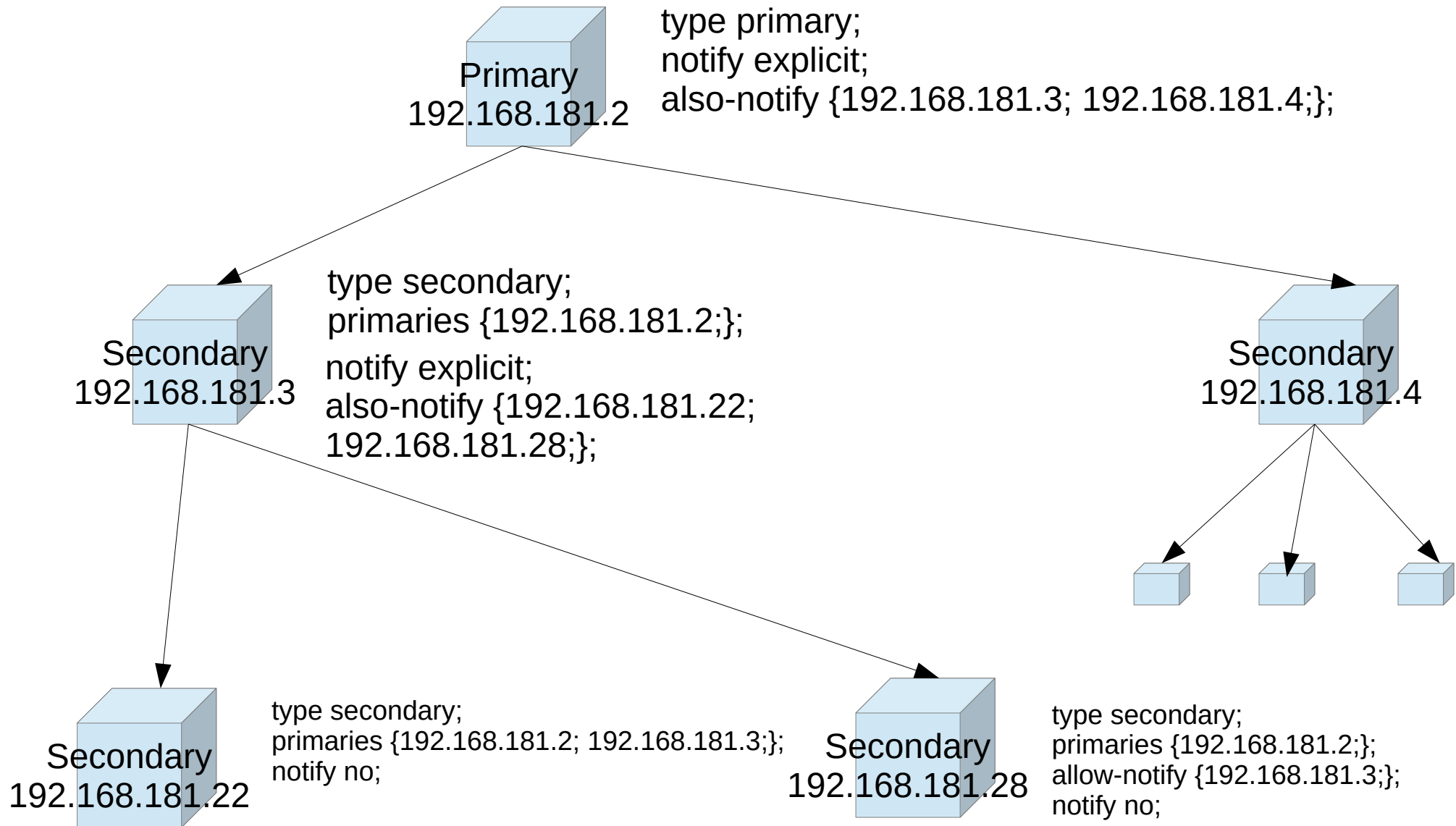
Configuring DNS - allow-update

- allow-update {address list;}; - The list of servers that are allowed to make dynamic updates to the zone entry.
 - Really only belongs on the primary (and it will then pass those updates to the secondaries).
 - Dynamic updates, while convenient, are less secure.
 - We'll see how to secure them better in OPS300. Since we can't secure them yet, do not use this.
 - Note: This is **not** necessary for normal updates between a primary and secondary server.

Sample Configuration



Conifguring DNS



Secondary DNS Zone Files

- The SOA record is very important when dealing with secondary servers.
- It determines how often it will check with the primary for updates (along with several other things).

Configuring DNS - Zones

- Every zone needs an SOA record.

```
zone IN SOA <primary name server> <email contact> (  
    Serial number  
    Refresh timer  
    Retry timer  
    Expire timer  
    Negative answer timer  
);
```

- *zone* is the zone this entry is for.
 - The @ is used to refer to the zone itself.
- *<primary name server>* is the name of the primary name server for this domain.
- *<email contact>* is the email address of the admin to contact if there are issues with this zone. Note: use . Instead of @

Configuring DNS - Zones

- Serial number is a whole number used to determine which version of the data this record represents.
- Secondaries will check against this number to determine if they need to get new data.
 - If the serial number on the primary is higher than the one they have in their data, they retrieve the new copy of the data.
 - Note that even when they receive a message from the primary telling them to update (if you set notify to yes on the primary), they will not update if their data has a higher serial number.

Configuring DNS - Zones

- Refresh, Retry, Expire, and Negative (aka minimum) are all TTL values, and can all be set using the short forms described there.
- Refresh – How often to check with the primary for updates.
- Retry timer – How long to wait before checking again if you could not contact the primary.
- Expire timer – How long to keep giving authoritative answers if you cannot contact the primary.
- Negative answer timer – How long clients should keep cached negative answers.

Configuring DNS - Zones

- You may also see SOA records on one line, without the ();

```
senecac.on.ca. 3D IN SOA ns.senecac.on.ca. its.senecacollege.ca. 12345 6h 30m 4w 2h
```

Is functionally identical to:

```
senecac.on.ca. 3D IN SOA ns.senecac.on.ca. its.senecacollege.ca. (  
12345  
6h  
30m  
4w  
2H  
);
```

- The (); just allow you to spread the data over several lines.
- ; allow you to include comments.

Summary

- This class we have examined how secondary DNS servers work.
 - How they copy data from the primary.
 - How they know when to copy data from the primary.